

The Marchand Index: A Cap-Adjusted Off-Ice Attention Quotient for NHL Players

Submitted to: Cascadia Symposium on Statistics in Sports (CASSIS), September 12, 2026. Format requested: Oral.

NHL front offices increasingly weigh off-ice fan attention — jersey sales, ticket demand, social engagement, viral moments — alongside on-ice production when valuing players, yet no public, reproducible metric isolates a player’s off-ice draw from their skill. We introduce the **Off-Ice Attention Quotient (OAQ)**: each player’s public-attention residual against a $K=10$ peer group of skill-equivalent NHLers. Cap-adjusting OAQ yields the **Marchand Index**, a per-dollar measure of off-ice value. Market amplification is controlled by reporting both raw (`OAQ_observed`) and team-baseline-stripped (`OAQ_portable`) variants. We validate the composite against three independent observable outcomes (NHL jersey list, All-Star fan vote, FA-signing event study), audit the LLM theme classifier against 300 hand-labeled comments (macro-F1 + Cohen’s κ), and pre-register four hypotheses before the model is run on production data.

1. Motivation

Public hockey analytics has comprehensively modeled on-ice value (xG, GAR, RAPM, Corsi-family). The off-ice side — what drives merchandise, ticket, sponsorship, and viewership demand — remains effectively unmodeled in public work. Front offices ask: *which players generate fan attention beyond what their production alone predicts, and which deliver that attention efficiently relative to cap hit?* We do not claim to measure revenue; we measure **public attention as a proxy for fan demand**, validated against three observable revenue-correlated outcomes. The metric is named after Brad Marchand: a high-skill player whose public salience and polarizing identity exceed what peer-matched production predicts — the canonical case the index is built to detect.

2. Method

Engagement composite. For all ~700 active NHLers — **including role and depth players, who are central to the thesis** — we compute volume-only signals from publicly available sources: a 12-month **Current Engagement Score (CES)** (weighted z-score over Wikipedia pageviews, Google Trends, Reddit mention/upvote volume, Instagram follower count, and additional platforms as data availability permits) and a career-to-date **Brand Depth Score (BDS)** (Wikipedia mean, jersey-list appearances, All-Star selections, service years, captaincy). Net sentiment and polarization are computed but stored as **separate output dimensions**, never folded into CES — doing so penalizes polarizing players (Marchand, Tkachuk, Reaves), precisely those the index is built to surface.

Peer matching. For each player P , identify $K=10$ nearest neighbors in standardized skill space (position, age, PPG, TOI/G, points/60 at 5v5, role band, NHL-adjusted production) using Mahalanobis distance with cohort-specific scaling. Players whose median peer distance exceeds the 75th-percentile of NHLer-pair distances receive a `match_quality=low` flag.

OAQ — observed and portable. $OAQ_observed(P) = engagement_raw(P) - engagement_raw_{K=10\ peers}$. **OAQ_portable subtracts the player’s team-market baseline before peer comparison** — the attention that would travel with the player if signed elsewhere. Both are reported. `OAQ_portable` is the headline metric because it survives the strongest objection (*“isn’t Marner just famous because he plays in Toronto?”*).

Marchand Index. $MI = OAQ_portable / cap_hit_M$. Per-player bootstrap 95% CIs (1,000 resamples of the player’s mention pool).

Theme decomposition (validated). An OpenRouter-hosted LLM classifies each Reddit comment into one of eight themes (`skill`, `fight`, `personality`, `fashion/style`, `off-ice life`, `controversy`, `charity/community`, `relationship/viral`). Audited against 300 stratified hand-labeled comments **before** any theme-based finding is reported: macro-F1 ≥ 0.60 , Cohen’s $\kappa \geq 0.55$ (ship-floor); 0.70 / 0.65 (target).

Why not just rank by the NHL jersey list? The jersey list is a coarse top-20 — a *validation target*, not the metric. OAQ is continuous across ~700 players, isolated from skill, and decomposable into themes.

3. Validation plan and pre-registration

Four independent rigor gates, each with a floor and a target:

#	Gate	Floor / Target
1	Spearman ρ (<code>OAQ_portable</code> top-50, NHL annual top-20 jersey list)	≥ 0.40 / ≥ 0.50
2	Spearman ρ (<code>OAQ_portable</code> , All-Star fan-vote share), 2022/2023/2024 ¹	≥ 0.45 / ≥ 0.55
3	DID regression: 60-day team-account follower delta ~ signed-player <code>OAQ_portable</code> + team FE + season FE + cap-hit + position, top-50 FA signings since 2020	directionally positive / $p < 0.05$
4	Theme classifier: macro-F1, Cohen’s κ on hand labels	0.60 / 0.55 (F1); 0.55 / 0.65 (κ)

¹ The 2024-25 NHL All-Star Game was replaced by the 4 Nations Face-Off; 2022/2023/2024 fan-vote totals only.

Pre-registration. Four hypotheses — polarization, cultural crossover (mentions outside hockey subreddits), viral-event persistence (≥ 6 -month attention decay), theme concentration (Herfindahl on theme shares) — are committed to a public repository **before** the model is run on production data. All four are reported with effect sizes + 95% bootstrap CIs and a `confirmed` / `disconfirmed` / `inconclusive` verdict, regardless of direction.

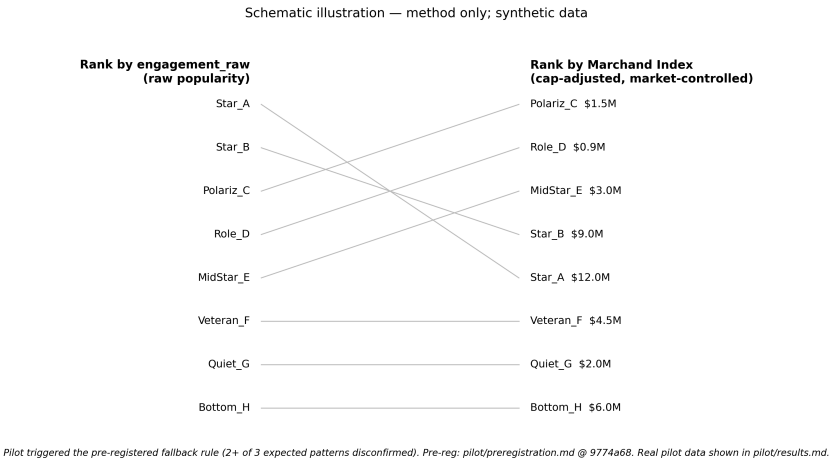
4. Preliminary result and sensitivity finding (worked-example pilot)

Pre-registered before fetch: `pilot/preregistration.md`, commit 9774a68 (2026-05-20). $N=14$ NHLers spanning skill (McDavid, MacKinnon, Makar, Draisaitl, Matthews, Kucherov), legacy (Crosby), polarization (Marchand, M. & B. Tkachuk), role-player archetype (Reaves), market-test (Marner — post-trade VGK), rising salience (Bedard, Hughes). $K=5$ restricted peer candidates within the 14; **not** validation of the full leaguewide $K=10$ method. Composite signals used: Wikipedia + Google Trends (Reddit and Instagram unavailable; pre-reg sentinel handling renormalized remaining weights).

The pilot pre-registered three falsifiable expected patterns. **Honest report: P3 confirmed (≥ 2 top-5 rank flips between raw-engagement ranking and Marchand-Index ranking); P1 and P2 disconfirmed.** Per pre-reg §11, ≥ 2 disconfirmations trigger a fallback to a schematic figure; the actual result is reported as a sensitivity finding rather than a headline number. **The disconfirmation is informative:** the pilot operationalized `team_market_baseline` as the mean rostered-player Wikipedia 12-month pageviews. This proxy conflates “team in a large

media market” with “team has other star players” — Florida’s roster mean is inflated by M. Tkachuk’s presence; San Jose’s by Macklin Celebrini’s. The mechanism therefore **over-corrects** for Marchand (FLA) and Reaves (SJS), exactly the limitation foreshadowed in §5. The full build replaces this proxy with a true media-market signal (team-account followers + arena attendance + market-population control) and applies the §3 gates.

The P3 confirmation matters: rank flips between raw popularity and the Marchand Index occur in the top-5 *even with* the over-correcting baseline, showing that the cap- and market-adjustment dimension is doing real work and is **not** a re-skin of popularity. A schematic side-by-side rank diagram appears below; the actual pilot CSV, pre-registration document, amendment log, and pattern-evaluation script are released alongside this submission.



5. Contribution and honest limits

Contribution. To our knowledge, the first publicly reproducible, peer-matched, market-controlled, cap-adjusted off-ice attention metric for NHL players, with a hand-validated LLM theme decomposition and pre-registered hypotheses. The full-build universe is all ~700 active NHLers; surfacing high-OAQ **role and depth players** (the Reaves archetype) is a primary intended finding. The framework generalizes to any league with public engagement signals.

Limits accepted. (1) Attention proxies revenue; the dollar denominator is not ours to assert. (2) Market control is approximated — team_market_baseline does not perfectly separate “has stars” from “big media market” (see §4). (3) X / Twitter engagement is unavailable (free API removed); follower-count snapshots only. (4) Goalies excluded from the headline analysis — peer matching breaks down. (5) Stars and rookies have few good peers; match-quality flags are shipped per-player. (6) The N=14 pilot is a worked example; leaguewide K=10 results are post-submission.

Reproducibility. Submission package: pilot pre-registration (committed prior to fetch), pilot scripts, pilot CSV, public repository link. Hand-labeled classifier validation set is a Wk 5 deliverable released later.

Selected references. Vollman R., *Hockey Abstract* (NHLe). Bacon P. (updated NHLe). Davis J., *Pick224* (pGPS peer-matching). Imbens G., Rubin D. (2015), *Causal Inference for Statistics, Social, and Biomedical Sciences* (matching estimators). Reddit / Wikimedia REST / pytrends API documentation.